

Vladimir Vekhov

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EDUCATION

Moscow Institute of Physics and Technology (MIPT)

Moscow, Russia

Bachelor's Degree in Applied Mathematics and Physics

2023 – 2027

GPA: 7 / 10

Core Courses: Linear Algebra (10/10), Differential Equations (8/10), Computational Mathematics (8/10), Physics (10/10), Algorithms (8/10), Microcontrollers (10/10), Digital Electronics (8/10).

EXPERIENCE

Institute for System Programming of the Russian Academy of Sciences

June – October 2024

Compiler Technology Department

Developed a signature generation algorithm (sigbox) for automated code analysis using C++; studied internet protocols and network architecture for traffic analysis.

Tarantool (VK)

October 2024 – July 2025

Database Kernel Developer

Implemented pagination support for the R-tree index in the Tarantool in-memory database kernel; worked with C and Lua, gaining deep understanding of database architecture and indexing.

Bureau 1440

July 2025 – Present

Satellite Control Systems Department

Developed and tested a software agent for satellite management; designed a space-radiation-resistant database and implemented an error analysis system.

PROJECTS

[My Football Programming Language](#) | *C, Recursive Descent Parser, CPU Simulator*

- Created a programming language using recursive descent parsing.
- Built a compiler supporting tokenization, abstract syntax tree creation, and assembly.

[RnD: Integration RS-algorithm in SQLite db](#) | *C++, meson, Reed Solomon, Sqliite orm*

- Sql Database Interaction Library integrated with Reed Solomon algorithm.
- Used for space-radiation-resistant database with an error analysis system.

[Telegram Catalog app](#) | *Python, fastapi, Docker, rec system*

- A Telegram application featuring smart category-based search, a recommendation system, an asynchronous interface, and a user-friendly design.
- I led a team of four programmers, managed the architecture, handled CI/CD and promoted the project.

[Differentiator](#) | *C, Maths, Binary Tree, Graphviz*

- Developed a tool for differentiating and simplifying mathematical expressions.
- Implemented a graphical tree representation using Graphviz.

[Physics laboratory assignments and research projects](#) | *Mechanics, Optics, Thermodynamics, Electricity*

- A repository containing my research projects: programs for calculating uncertainties in laboratory experiments, laboratory assignments.
- Measuring the specific charge of the electron and more.

[My Processor, RVI32 Processor](#) | *C, C++, Assembly, ISA*

- Created a processor and assembler supporting stack, RAM, and registers.
- 2 types: stack-based CPU with my own instructions and RVI32 simulator CPU.

[Game Shooter](#) | *C++, SFML, Sockets, threads*

- Top-down shooter - simple 2D video game with client-server api, graphics and game engine on SFML.

[Intersection of triangles](#) | *C++, BVH tree, Linear Algebra, CI*

- This program detects intersections between triangles in 3D space.
- Implemented a BVH tree for optimisation.

All blue links are clickable and lead to my projects on GitHub.

AWARDS

Physics Olympiad Winner: Rosatom, Kurchatov, St. Petersburg, Engineering Olympiads.

ABOUT ME

- Languages & Technologies: C, C++, Python, Lua, SQL, Bash, Assembly
- Frameworks & Tools: SolidWorks, CMake, Meson, Docker, Git, Gitlab, FastAPI, Wireshark, Graphviz
- Concepts: OOP, Data Structures & Algorithms, Network Protocols (TCP/UDP), RISC-V, OS (Linux), Compiler Design, Optimisations
- Teaching Experience: Programming mentor for MIPT C course (1 year, under I. Dedinsky) and Physics teacher for children.
- Soft Skills: Strong teamwork, brainstorming skills and communication, led a 4-person developer team.
- Languages: Russian (native), English (B2, technical reading/writing), Italian and German (A1/A2).
- Hobbies: Football, running, guitar, microcontrollers, reading.